7.2 Area of Triangles – Part 2

Lesson Introduction

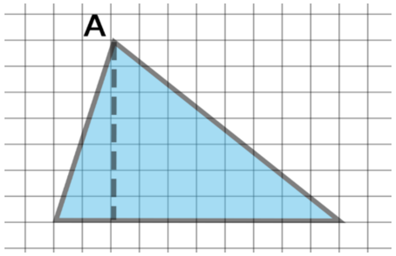
Benchmark	MA.6.GR.2.1 Derive a formula for the area of a right triangle using a rectangle. Apply a formula to find the area of a triangle.	Learning Target	• I can apply a formula to find the area of triangles.

	Building On	Working Toward
Benchmark Coherence	MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.	MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi.

Essential Questions	• Where is the base of a triangle in relation to the height? • How can you use the base and height of a triangle to find the area? • How can you find the height of an acute or obtuse triangle?
Lesson Narrative	In this lesson, students use the formula for the area of a triangle to solve for the area of triangles. The lesson starts by making sure students can correctly identify the base and height of triangles. Once students have an understanding on how to find the base and height, they substitute values into the formula to solve for the area of a triangle.

Component Summary	7.2.1 Warm-Up (~5 min.)	Find the area of a triangle and rectangle and describe their relationship *(SE p. 303)*	7.2.3 Guided Practice (10 min.)	Identify base and height from given measurements on a triangle and calculate the area *(SE p. 305)*
	7.2.2 Guided Instruction (20-25 min.)	Find base and height by finding perpendicular lines in the given triangle *(SE p. 303)*	7.2.4 Your Turn! (10 min.)	Identify base and height from given measurements on a triangle and calculate the area *(SE p. 306)*
	7.2.5 Wrap-Up (~5 min.)	Identify base and height from an obtuse triangle to calculate area *(SE p. 307)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Base • Height • Area of a triangle Additional Terms: • Perpendicular	• (Optional) Straight edge • (Optional) Graph paper • Four-Function Calculator	N/A

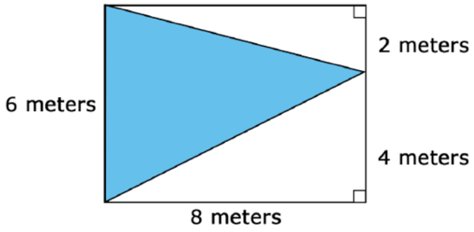
	Common Misconceptions	Things to Consider
Teacher Tips	• Students may identify lines that are perpendicular to the height rather than the base (like answer choice c in 7.2.3 Guided Practice). • Students may incorrectly identify a non-perpendicular side of a triangle as the base or height. • Students may get stuck while substituting values into the area of a triangle formula.	• Take time to discuss how base is not always the bottom and height is not always the vertical side. • When given more than two measurements on a triangle, encourage students to label the height and base for each problem. • When needing to draw the height outside of an obtuse triangle, remember that the base does extend to the drawn line. • Consider allowing students to have access to four-function calculators.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Clarify, Critique, Correct Consider utilizing a clarify, critique, correct learning routine for 7.2.3 Guided Practice. Provide students with the following statement: The base of a triangle is the bottom line, and the height is the longest line. Prompt students to discuss the problems with the given statement and how they could rewrite.	**MA.K12.MTR.6.1: Reasonableness** This lesson provides multiple opportunities for students to check their calculations and determine the reasonableness of their solutions. Encourage students to review their work as they use the formula to find the areas of the triangles. This will reinforce the importance of checking their work as they progress through tasks and encourage mathematical precision.
Differentiate Instruction	Utilize graphing paper or similar resources as scaffolding for students to recognize the perpendicular relationship between the base and the height. By drawing the triangle on the grid the student can use the gridlines on a coordinate plane to draw the height Consider modeling this approach in 7.2.2 Guided Instruction and prompting students to use this method during 7.2.3 Guided Practice. 	

Lesson Notes:

7.2.1 Warm-Up: Bell Work

Component Overview: *In this Warm-Up, students find the areas of a triangle and a rectangle using dimensions from a diagram of a triangle inscribed in a rectangle.*

7.2.1 Warm-Up: Bell Work



1. What is the area of the rectangle? Include units.
   48 m^2

2. What is the area of the shaded triangle? Include units.
   24 m^2

3. Explain the relationship between the area of a rectangle and triangle that have the same height and base.
   The area of a triangle is half the area of a rectangle with the same height and base.

Use the bell work to inform instruction and student misconceptions. Consider reviewing student work to notice if students used the formula for question 2 or if they found ½ of the area of the rectangle. If students struggled, consider using small group instruction to help students with their misconceptions.

7.2.2 Guided Instruction: Find the Area of a Triangle

Component Overview: *In this Guided Instruction, students explain how base and height are related within a triangle. Students use the given base and height to calculate the area of a triangle.*

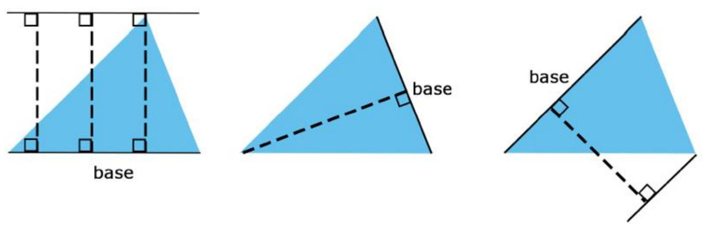
7.2.2 Guided Instruction: Find the Area of a Triangle

To find the area of any triangle, identify a base of the triangle and its corresponding height.

Any of the three sides of a triangle can be the **base.** The term "base" can refer to the side of a triangle and also its measurement.

The corresponding **height** of a triangle is the length of the perpendicular segment drawn from a base to its opposite vertex. The opposite vertex is the vertex of the triangle that is *not* an endpoint of the base.

Examples of bases and their corresponding heights on copies of the same triangle are shown. Each dashed line segment represents the height of the triangle.



1. In each example the height is
   - ○ perpendicular
   - ○ opposite

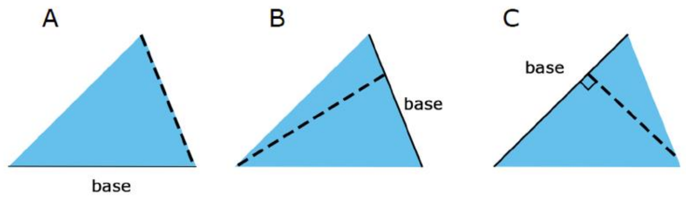
   to the base.

This Guided Instruction is intentionally scaffolded to solidify student understanding before applying the formula to find the area of a triangle. Consider having a student read each blue box and go through each example provided. Encourage students to ask questions as the class reviews each example. Guide the discussion with questions such as:

- *What does "perpendicular" mean?*
- *What is the difference between base and height?*
- *Why do some of the heights in the triangles go outside out of the triangles?*

7.2.2 Guided Instruction: Find the Area of a Triangle (continued)

2. The side used for the base is labeled on three triangles below. Complete the table to explain why each of the dashed lines do not represent the height of each triangle.



Triangle	Explanation
Triangle A	*The base is not perpendicular to the height.*
Triangle B	*The base does not form a* $90°$ *angle with the height.*
Triangle C	*The height does not reach the highest point of the triangle.*

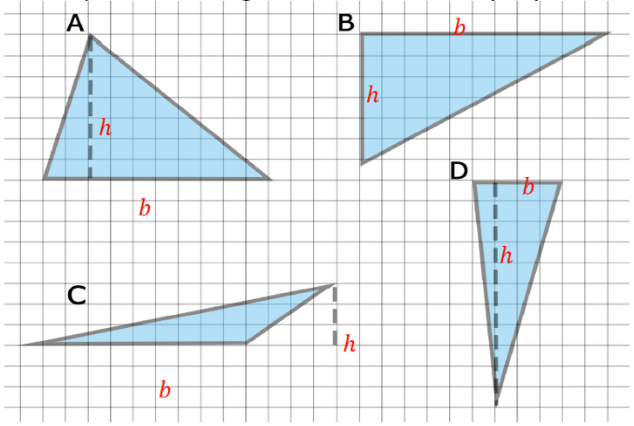
Area of a Triangle $= \frac{1}{2}bh$,
where b is the base and h is the height.

Question 2 intentionally shows bases using different orientations. Make it a practice when working with students to vary the position of the base. Any side of a triangle can be called the base if there is a perpendicular that is drawn from the base to its opposite vertex to represent the height.

Consider having students who struggle visually to draw and cut out separate triangles with lines drawn like each option so that they can rotate each one to have a better visual when the base is on the bottom.

7.2.2 Guided Instruction: Find the Area of a Triangle (continued)

3. Four triangles are shown on the grid. The side length of each square on the grid is 1 centimeter (cm).



a. For each triangle, label the line segment that is drawn to represent the height, h. Then, label the side that is the base, b. *Answer shown on grid.*

b. In the table, record the measurement for the base and corresponding height for each triangle. Then, find the area of each triangle. Each square on the grid represents 1 centimeter (cm) by 1 cm.

Triangle	Base (cm)	Height (cm)	Area (cm^2)
Any Triangle	b	h	$\frac{1}{2}bh$
Triangle A	10 *cm*	7 *cm*	35 cm^2
Triangle B	11 *cm*	6 *cm*	33 cm^2
Triangle C	10 *cm*	3 *cm*	15 cm^2
Triangle D	4 *cm*	11 *cm*	22 cm^2

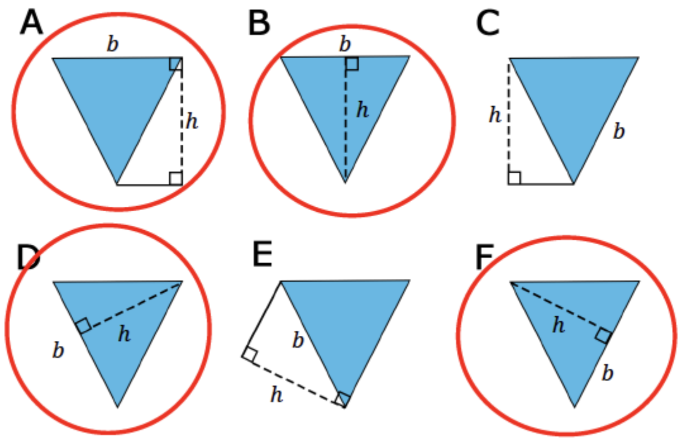
For question 3b consider completing the table for Triangle A then releasing students to complete the table while circulating listening for student misconceptions.

7.2.3 Guided Practice: Find the Area of a Triangle

Component Overview: *In this Guided Practice, students work with a partner to identify correctly labeled base and height within a triangle and use that base and height to find the area of a triangle.*

7.2.3 Guided Practice: Find the Area of a Triangle

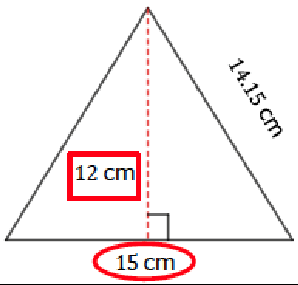
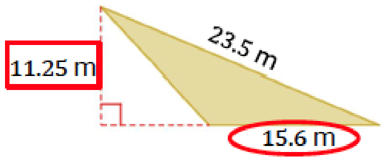
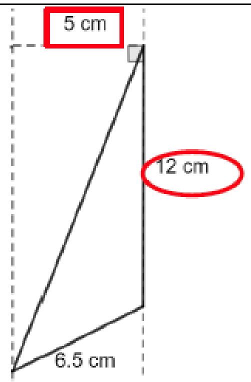
1. Discuss with your partner which drawings correctly label base, b, and its corresponding height, h, represented by the dashed line segment. Circle all the correct drawings. *Answer shown below.*



Clarify, Critique, Correct
Have students read the following sentence: "The base of a triangle is the bottom line, and the height is the longest line of a triangle."
Have students discuss what the problems are with this statement and how they could rewrite. If multiple students are getting different answers, position them to turn and talk about why they think they are right with their partner and foster a discussion for students to arrive at the same answer.

7.2.3 Guided Practice: Find the Area of a Triangle (continued)

2. For each triangle below, circle the base measurement and put a square around the height measurement. Then, use the base and height to find the area of each triangle. Show your work. Include units.

	Area = 90 cm^2
	Area = 87.75 m^2
	Area = 425.25 mm^2
	Area = 30 cm^2

Consider having students first circle the base and square the height and then come back together as a class to check in before finding the area. Ask questions such as:
What helped you determine the height?
What helped you determine the base?

Lead students to understand that the marking of the right angle can be helpful in denoting which side is the base.

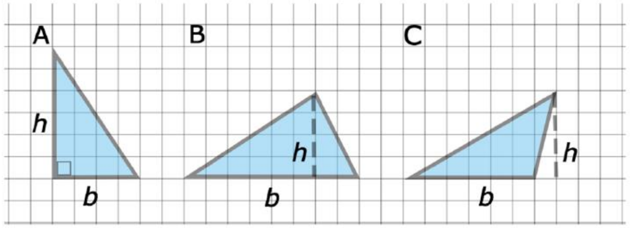
Consider having students write the units for the area on the answer line before they solve the problem to not forget or write the incorrect ones.

7.2.4 Your Turn!

Component Overview: *In this Collaboration, students use the given height and base of each triangle to find the area of each triangle. Students decipher which measurements are used for base and height and which are not.*

7.2.4 Your Turn!

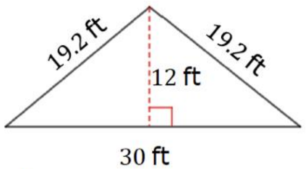
1. For each triangle, a base and its corresponding height are labeled. Complete the table by finding the area of each triangle. The side length of each square on the grid is 1 inch (in.).



Triangle	Area (in^2)
A	12 in^2
B	16 in^2
C	12 in^2

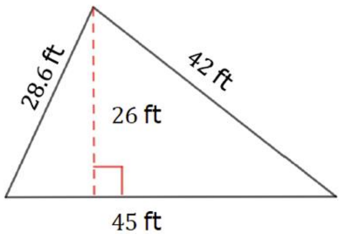
2. Find the area of each triangle. Include units.

a.



Area = 180 ft^2

b.



Area = 585 ft^2

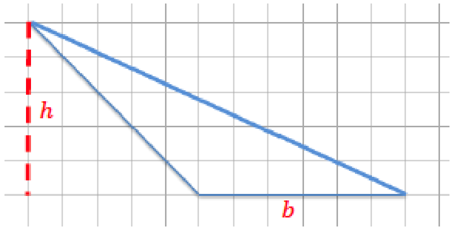
Relate measuring the height of an object to when you measure your own height. Relate what perpendicular lines look like by having students stand and describe their bodies' relationships to the floor. Having students use their arms to show perpendicular lines will engage kinesthetic learners.

7.2.5 Wrap-Up: Ticket Out the Door

Component Overview: *In this Wrap-Up, students find the base and height measurements of a triangle and calculate the area.*

7.2.5 Wrap-Up: Ticket Out the Door

1. A diagram of a triangle is shown on a grid. The side length of each square on the grid represents 1 foot (ft).



   a. Label a base and its corresponding height for the triangle.
   Shown on image.

   b. Find the area of the triangle. Include units.
   15 ft^2

Use the bell work and the ticket out the door to inform you of student progression on misconceptions. If students struggled, consider using small group instruction to help students with their misconceptions.